

Final Assignment KSSC Report

Saturation and Hallucination Analysis on RAG

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1. Method

RAG Architecture

The system follows a standard RAG architecture: a dense-embedding retriever fetches the top-K relevant documents from a corpus, which are then injected into a prompt fed to a generator LLM to produce the final answer.

Dataset

IDK-MRC (rifkiaputri/idk-mrc), an Indonesian machine reading comprehension dataset containing both answerable and unanswerable questions. Evaluation is performed on the validation split using a fixed sample of 150 questions (~105 answerable, ~45 unanswerable), with random seed 42 for reproducibility.

Selected Models

- **Embedding / Retriever:** `intfloat/multilingual-e5-base` (sentence-transformers), chosen for its strong multilingual retrieval performance.
- **Vector Store:** FAISS (IndexFlatIP) with cosine similarity (inner product on normalized vectors).
- **Generator LLM:** Qwen/Qwen2.5-3B-Instruct, loaded with 4-bit quantization (NF4 via BitsAndBytes) to fit within a free-tier T4 GPU.

Prompt Design

The prompt is written in Indonesian to match the dataset language, which leads to outputs that align more closely with the gold answers and yields better ROUGE, BLEU, and BERTScore. The prompt uses a two-example few-shot structure: one answerable example and one unanswerable example. The second example explicitly teaches the model to respond with "Tidak tahu" (I don't know) when the answer cannot be found in any of the provided documents which serve as an abstention mechanism to suppress hallucination on unanswerable questions. The model is also required to cite the source document number in every response.

Retrieval Granularity (Two Experiments)

We run the experiment under two retrieval configurations. In **Experiment 1 (full paragraphs)**, each document is a whole paragraph (358 documents). In **Experiment 2 (chunked)**, each paragraph is split into ~ 45 -word chunks with a 12-word overlap (884 chunks), so the answer sits in one specific chunk and retrieval becomes a finer-grained task. Both experiments use identical components and prompts; only document granularity differs.

2. Experimental Results

The experiment was conducted across four values of K (1, 3, 5, 10) on a fixed sample of 150 questions from the IDK-MRC validation split. Recall@ K reached approximately 0.990 at $K=1$ and plateaued at 1.000 from $K=3$ onward. BERTScore-F1 remained consistently around 0.80 across all K values, while ROUGE-L ranged between 0.55 and 0.56, and BLEU ranged between 0.33 and 0.35. Citation Accuracy fluctuated between 0.858 and 0.879 with no strong directional trend. The hallucination rate on unanswerable questions ranged from 0.733 at $K=1$ to a high of 0.800 at $K=3$ and $K=10$, with a slight dip to 0.778 at $K=5$. Image 1 shows the plots of the values, while Table 1 gives the exact values found.

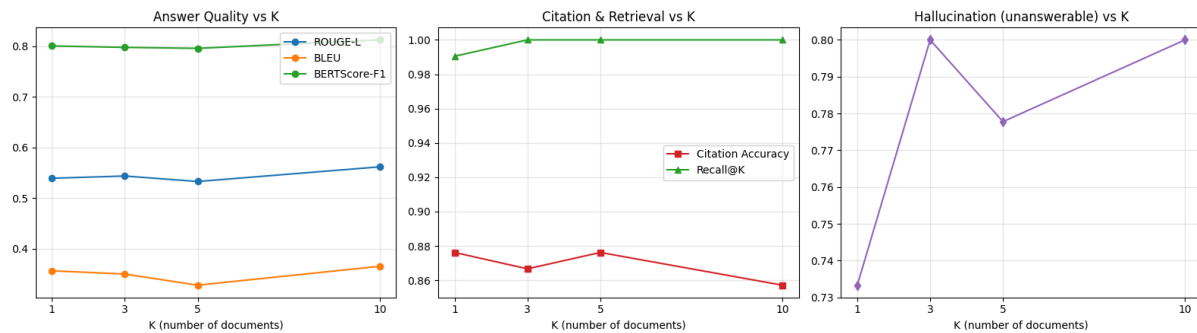


Image 1 (3 different graphs: Answer Quality vs K, Citation & Retrieval vs K, Hallucination vs K)

K	ROUGE-L	BLEU	BERTScore-F1	Citation Accuracy	Recall@K	Hallucination Rate (unans)
1	0.56	0.34	0.80	0.879	0.990	0.7333
3	0.56	0.33	0.80	0.864	1.000	0.800
5	0.55	0.33	0.79	0.879	1.000	0.778
10	0.56	0.35	0.80	0.858	1.000	0.800

Table 1 (3 different graphs: Answer Quality vs K, Citation & Retrieval vs K, Hallucination vs K)

The same experiment was repeated under the chunked Experiment (Experiment 2). Here Recall@ K rises from 0.895 at $K=1$ to 0.971 at $K=3$, then 0.990 at $K=5$ and $K=10$, and BLEU increases steadily from 0.233 to 0.345 as K grows. Citation Accuracy decreases from 0.867 ($K=1$) to 0.762 ($K=10$), and the hallucination rate rises from 0.556 to 0.622. Table 2 gives the exact values.

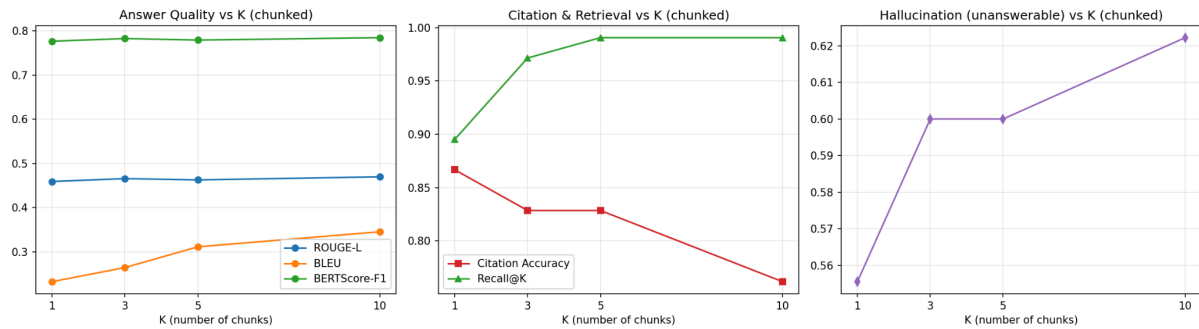


Image 2 (3 different graphs: Answer Quality vs K, Citation & Retrieval vs K, Hallucination vs K for Experiment 2)

K	ROUGE-L	BLEU	BERTScore-F1	Citation Accuracy	Recall@K	Hallucination (unans)
1	0.459	0.233	0.776	0.867	0.895	0.556
3	0.466	0.264	0.782	0.829	0.971	0.600
5	0.463	0.311	0.779	0.829	0.990	0.600
10	0.470	0.345	0.784	0.762	0.990	0.622

Table 2 (3 different graphs: Answer Quality vs K, Citation & Retrieval vs K, Hallucination vs K for experiment 2)

3. Analysis & Discussion

Performance Saturation at a Certain K

Recall@K is expected to increase as K grows, since retrieving more documents raises the likelihood that the gold document is included in the context. However, answer quality metrics (ROUGE-L, BLEU, BERTScore) tend to peak at a specific K value and then plateau or decline. This pattern reveals a saturation point that can originate from two different places:

- If Recall@K continues to rise while answer quality stagnates, the saturation originates from the generator. The model fails to extract the relevant information from an increasingly long and noisy context.
- If Recall@K itself plateaus early, the saturation originates from the retriever. The embedding model can no longer distinguish more relevant documents beyond what was already retrieved at a lower K.

Large contexts (high K) can also confuse the LLM even when the correct document is present, a phenomenon sometimes referred to as "lost in the middle," where the model disproportionately attends to the beginning and end of the context and overlooks relevant passages in between.

The Role Of Retrieval Granularity

Comparing the two experiments makes the saturation point clearer. In experiment 1, Recall@1 is already 0.990, so the gold document is almost always retrieved with a single document and adding more K cannot improve anything, which is why answer quality stays flat. The saturation is therefore immediate, and the flat quality reflects a generation-side limit, not retrieval. In experiment 2, where the answer sits in one specific chunk among 884, retrieval becomes a real variable: Recall@K rises and plateaus around K=5, and BLEU rises in step. This is the expected saturation curve, and it only appears once retrieval is hard enough to matter.

Citation Hallucination

In the chunked experiment, Citation Accuracy drops clearly as K grows (0.867 → 0.762). Because Recall@10 is about 0.99, the gold chunk is almost always present in the context, so these citation errors are generation-side: in one case the model answered "Kabupaten Banyuwangi" correctly but cited chunk [492] instead of the gold chunks [685, 686, 687], a right answer with a wrong source.

Hallucination on Unanswerable Questions

The hallucination rate rises monotonically with K in the chunked experiment (0.556 → 0.622), meaning more context pushes the model to answer questions it should refuse. Notably, even with explicit grounding instructions and few-shot refusal examples, the 3B model still fabricates an answer on more than half of the unanswerable questions, showing that prompt-level grounding alone is not enough for a small generator.

Key Finding

At K=10 in the chunked experiment, Recall@K is at its highest (0.990) while Citation Accuracy is at its lowest (0.762) and hallucination at its highest (0.622). The correct chunk is in the context, yet the model still makes mistakes in citations and hallucinates more. This confirms that the high-K failures come from generation, not retrieval: adding context improves retrieval recall but lowers generation faithfulness.

Limitations

Abstention check is keyword-based, so refusals phrased in the model's own words (e.g. "the document does not mention...") are counted as hallucinations; the reported hallucination rate is therefore a slight overestimate. BLEU is also low because IDK-MRC answers are short, making ROUGE-L and BERTScore more informative here. Finally, we use a single generator and 150 questions, so results may differ at a larger scale.

4. Conclusion

RAG saturation is only visible when retrieval is challenging: full-paragraph retrieval saturates instantly at K=1, while chunked retrieval shows a saturation curve that levels off around K=5. Increasing K lowers generation faithfulness because citation accuracy falls and hallucination

risks, even when retrieval is at its best, so the high-K failures come from generation, not retrieval. Choosing K thus means balancing retrieval coverage against hallucination risk, and prompt-level grounding alone is insufficient for a small generator.

Work Division

- La Tania Nur Tenka (23/511404/PA/21787): Retrieval (Recall@K)
- Rr. Garzetta Aleyda Harimurti (23/511422/PA/21793): Generation & citation (Citation Accuracy)
- Leilani Fitria Salimah Hadiwibowo: Evaluation (Answer Quality + Hallucination)